TRSANet: A Remote Sensing Deep Learning Model for Water Body Change Detection Based on Time-Reversal Semantic Asymmetry

Jiasheng Li, *Graduate Student Member, IEEE*, Chenhao Jin, Yao Shen, and Weixiang Ye

***Abstract*— The goal of change detection is to identify modifications in dual-temporal remote sensing (RS) images, which reveal dynamic changes in specific areas of interest. Specifically, the detection of water body changes via RS is critical for monitoring environmental shifts, such as droughts and floods. However, RS methods that utilize deep learning (DL) to detect water changes face two challenges: capturing the unique spatial-temporal features of changes in water bodies and effectively modeling temporal features and spatiotemporal coupling. Drawing inspiration from the theory of time-reversal asymmetry in physics and considering the temporal characteristics of RS water bodies, we propose the time-reversal semantic asymmetry module to extract essential temporal-spatial features more effectively. Consequently, this article presents the design of a model specifically tailored for detecting changes in water bodies, termed time-reversal semantic asymmetry network (TRSANet). TRSANet outperforms other state-of-the-art methods by not only having fewer parameters but also achieving the highest $F1$ score and IOU (91.71% and 84.69%, respectively) on the Water-CD dataset for water body change detection. Additionally, TRSANet demonstrates remarkable cross-domain adaptability by attaining the highest accuracy on the LEVIR-CD dataset for building change detection. In the future, TRSANet will also offer methodological support for urban planning and the establishment of water body conservation areas.**

***Index Terms*— Remote sensing (RS), time-reversal semantic asymmetry, water body change detection.**

I. Introduction

CHANGE detection, a pivotal aspect of remote sensing (RS), aims to identify alterations in specific areas by comparing images captured at different times [1]. This technique has profound implications for environmental monitoring [2], disaster detection and assessment [3], [4], urban planning [5], and resource management [6]. Water body change detection is crucial for dynamic monitoring and real-time surveillance of areas affected by droughts and floods [7]. Automated change detection technologies, which offer superior results with reduced manpower and time costs, are thus attracting increasing attention [8].

Traditional change detection methods primarily rely on spectral information in RS images to identify land cover changes. Techniques such as principal component analysis (PCA) [9], multivariate alteration detection (MAD) [10], change vector analysis (CVA) [11], and Markov Random Fields are commonly used. However, these methods necessitate the selection of optimal thresholds for decision-making, which introduces additional workload. Machine learning algorithms, including support vector machines (SVMs) [12] and Random Forests [13], have also been widely applied in change detection. Nonetheless, these methods heavily depend on manually designed features based on prior domain knowledge, rendering them susceptible to seasonal variations, lighting conditions, and satellite sensor characteristics. This dependency undermines their robustness and poses challenges to the generalization ability of the models.

Recent advances in big data and increased computational power have significantly enhanced the capabilities of deep learning (DL) in RS applications, particularly in change and target detection [15], [16]. Convolutional neural network (CNN), with their strong feature extraction abilities, have been effectively utilized to autonomously identify diverse features from RS data. Additionally, the integration of Transformers [17] has further advanced the field of change detection [18], [19]. Unlike methods that use a single-image input for land cover classification [20], change detection often employs complex architectures such as dual-stream Siamese networks for extracting features from dual-temporal images [21] or single-stream networks incorporating an early fusion strategy [20]. For example, Zhu et al. [22] introduced the Siam-GL framework, a novel semantic change detection framework designed to address issues of sample imbalance and background influence. Recent innovations in DL for change detection focus on establishing spatial connections among different temporal images. Fang et al. [23] introduced SNUNet, which utilizes densely multiscale semantic features for better fusion, enhancing the effectiveness of feature extraction. Du et al. [24] proposed DSFA for multitemporal

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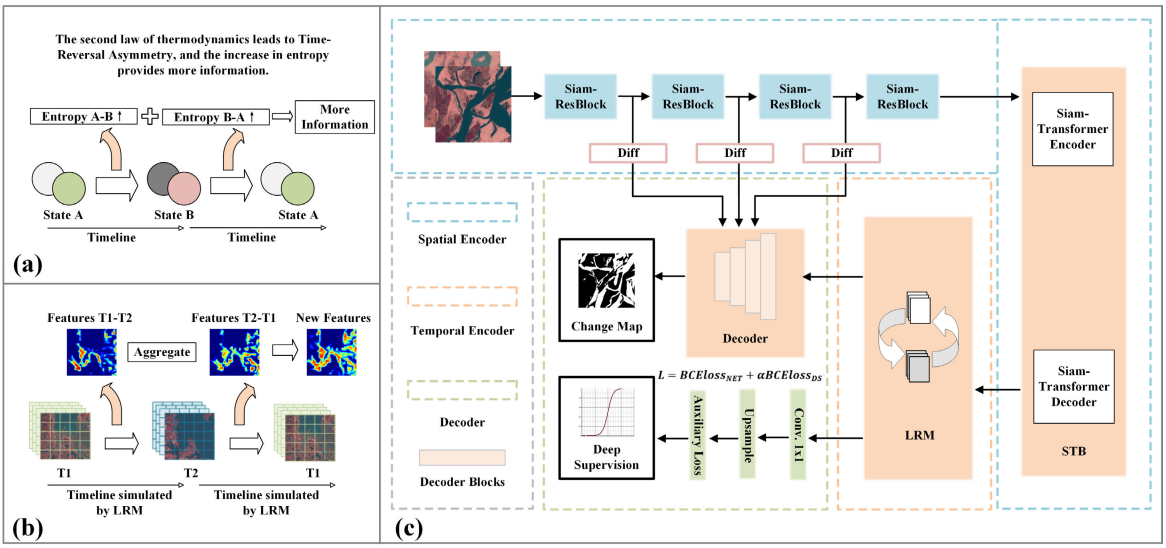


Fig. 1. (a) Basic principles of time-reversal asymmetry from physics. The second law of thermodynamics leads to time-reversal asymmetry, and the increase in entropy provides more information. (b) Basic principles of time-reversal semantic asymmetry. Due to the periodic changes in water bodies, LRM simulates bidirectional timelines by directly considering a past moment T1 as the future moment of T2, extracting and aggregating the time-reversal semantic asymmetric semantic features in the bidirectional changes. (c) Overall framework of TRSANet. The encoder takes the bi-temporal image pair as input, employing the STB to extract high-level semantic features into the LRM. The enhanced LRM captures bidirectional asymmetric spatiotemporal information. The decoder utilizes the output of LRM alongside multilevel difference features extracted by the CNN backbone. Additionally, the DS of the LRM output enhances attention to evolving components.

RS image change detection, employing symmetric deep networks and slow feature analysis to effectively suppress unchanged components and highlight changed ones. Similarly, Ding et al. [25] developed DSANet, a dual-branch end-to-end network that incorporates spatial attention (SA) mechanisms to focus learning on areas of change. Despite these advancements, many current change detection networks, particularly those using Siamese architectures, face challenges in efficiently processing temporal interactions among different temporal images. Traditional DL methods for feature extraction often fail to fully exploit available temporal information, simplifying the change detection process or focusing solely on unidirectional changes, thus limiting the effectiveness of feature extraction and information fusion.

In addressing these challenges, our approach is inspired by the concept of "Time-Reversal Asymmetry" from physics, which posits that change processes exhibit directionality and are asymmetric in both forward and reverse temporal contexts. This principle suggests that the transition of an object from state A to state B is distinct from its reversal, as illustrated in Fig. 1(a). This is consistent with the second law of thermodynamics, where both transitions are associated with an increase in entropy, adding a layer of informational depth. Additionally, changes in water bodies often display periodicity, indicating that regions of past water bodies may be predictive of future states. A model that only learns unidirectional increases or decreases in water bodies captures only partial information, which can hinder comprehensive learning.

Our conceptual model treats the transformation process of water bodies as asymmetric vectors, designed to capture richer time-reversal semantic asymmetric semantic features. Building on this concept, we introduce a novel change detection framework named TRSANet, specifically developed for detecting changes in water bodies. Our methodology employs a Siamese CNN backbone to derive multilevel spatial features from paired images. These features are further processed through a Siam-transformer block (STB) to extract high-level semantic features. Due to the periodic changes in water bodies, we utilize the Semantic Asymmetry property with Bi-ConvLSTM [27] by feeding the semantic features of image pairs in different sequences into our LSTM recycle module (LRM). This simulates bidirectional timelines by directly considering a past moment as a future moment, extracting and aggregating the asymmetric spatio-temporal features in the bidirectional changes, as depicted in Fig. 1(b). Additionally, the CNN backbone extracts multilevel difference features which are skip-connected to the decoder to integrate both temporal and spatial data. The model also incorporates deep supervision (DS) techniques to enhance the extraction of bidirectional spatio-temporal features, thereby improving the targeted learning efficacy of the LRM. The overall structure of TRSANet is depicted in Fig. 1(c).

The main contributions of this article can be summarized as follows:

1) We integrate the time-reversal asymmetry from physics into the change detection task, considering that additional asymmetric information and increased entropy are generated between dual-temporal image pairs during their mutual transitions. Guided by this concept, we designed the LRM module to extract and aggregate bi-spatiotemporal features.
2) Development of an efficient model named TRSANet, based on the LRM, specifically for detecting changes in water bodies: This model excels in extracting change features from dual-temporal water body images, effectively integrating temporal and spatial data.

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This integration allows for a more precise capture and understanding of changes within image pairs. We conducted extensive experiments on a water body change detection dataset we developed, which demonstrated the effectiveness and efficiency of TRSANet. Additionally, the model achieved state-of-the-art performance on publicly available building change detection datasets, highlighting its exceptional cross-domain adaptability. TRSANet is expected to make a significant contribution to the automatic monitoring of water body dynamics and disaster prevention.

II. Related Work

A. Water Body Change Detection

The extraction of water bodies and the detection of changes in RS imagery are core research topics in the field of RS. Insights from these investigations are crucial for critical applications like water resources monitoring, flood surveillance, and urban development. The automated extraction of water body change maps and the creation of long-term change graphs are essential for understanding the dynamic nature of surface water bodies.

Various methodologies have been used to extract water bodies from RS images, categorized into conventional and AI-powered techniques. The predominant conventional approach involves using RS indices, thresholds, and morphological operations [14]. In this method, appropriate thresholds are selected to distinguish water from nonwater in one or multiple spectral bands. Key indices include the normalized difference water index (NDWI) [28], modified NDWI (MNDWI) [29], and automated water index (AWI) [30]. These methods offer the advantages of simplicity and ease of use. The resulting extracted regions are then analyzed to detect changes in water bodies. However, these approaches require manually setting suitable thresholds, which increases labor and time costs and may result in less stable results when handling RS image pairs acquired under varying conditions.

In recent years, methods for detecting changes in water bodies have increasingly shifted toward machine learning and DL. Algorithms using machine learning classifiers [12], [13] achieve enhanced performance through feature engineering and learning processes. For instance, Peng et al. [31] employed a decision-tree-based multispectral index method to extract the surface water coverage area of Dongting Lake over multiple temporal phases, analyzing water body changes from 1975 to 2019. Li et al. [32] used an unsupervised approach within a Bayesian network fusion framework to detect urban inundation. Nevertheless, these approaches are heavily contingent on handcrafted features with specific domain knowledge.

In the context of water bodies, the application of DL is predominantly focused on water body segmentation tasks. Dui et al. [26] introduced the E-Unet, which incorporates multiscale decoder (MSD) path modules and multiscale processing (MSP) modules, enhancing the ability to discern multiscale information and thereby improving segmentation performance in urban areas with small rivers. Concurrently, DL has shown exceptional utility in detecting changes in water bodies, significantly enhancing the accuracy of traditional methods. Bai et al. [33] utilized weakly supervised datasets to train a DL model, effectively identifying permanent and temporary water bodies during flood events. Wu et al. [34] introduced the MS-Deeplab, a water detection engine encapsulating polarimetric information from synthetic aperture radar (SAR) imagery, accompanied by a dual-stream feature extraction substrate and a multitiered feature amalgamation unit. They utilized multitemporal information to detect flood-affected areas. Jiang et al. [35] introduced an SAR image water surface mapping method based on Attention-UNet3+ to map water body changes in the Poyang Lake region. The advantages of these methods include the availability of extensive datasets for training. Despite the advancements in DL techniques for water body monitoring, they are still limited to using segmentation to monitor changes in water bodies, thus overlooking the temporal dynamics inherent in water body changes. In other words, the temporal characteristics of water body changes remain largely unexplored. This forms one motivation for our work.

B. Modeling of Time in Change Detection

In contrast to other image-processing tasks, dual-temporal images inherently exhibit a strong spatiotemporal correlation. Typical strategies for handling temporal features include channel concatenation and feature differencing. Scholars have also explored extracting crucial spatial differentiators [36] and applying style transference [37] to mitigate pseudo-alterations in change detection tasks. Temporally, investigations have leaned toward recurrent neural network (RNN) [38], 3-D convolutions [39], and spatiotemporal attention mechanisms [40]. While 3-D CNNs are commonly utilized for video processing within dual-stream networks, some researchers interpolate dual-temporal images, generate pseudo-video frames, and employ 3-D CNNs to extract temporal information from these pseudo-frames. Despite the excellent temporal extraction capabilities of video-based 3-D CNNs, they still encounter challenges associated with high computational complexity.

In the model based on RNN, Mou et al. [41] pioneered the amalgamation of CNNs with RNNs for temporal feature interpretation in change detection, examining three disparate RNN architectures (RNN [38], GRU [42], LSTM [27]) and demonstrated that LSTM is the most suitable for CD tasks. Papadomanolaki et al. [43] employed LSTM blocks with integrated convolutional layers combined with Unet to model unidirectional temporal changes in image pairs. While these methods effectively handle buildings and urban areas with continuous growth trends by considering changes as unidirectional or directionless, they overlook an important aspect. Buildings and urban areas with continuous growth trends also include their reduction data, it simply indicates that the growth rate is greater than the reduction rate. For targets with clear periodic changes, such as water bodies, the growth rate is almost the same as the reduction rate. In these cases, past water body regions may reappear in the future, making it more crucial to view these changes as a reciprocal process. Adopting a "Time-Reversal Semantic Asymmetry" approach becomes more necessary in these instances. In this context,

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Algorithm 1 Inference of TRSANet

Input: I = {(I_1, I_2)} (a pair of RS water body images)
Output: M (a change detection prediction mask)

// Step1: Extracting semantic features by CNN
// backbone and Siamese Transformer Block
for i **in** {1,2} **do**
  $X^i = Backbone_CNN(I^i)$
  $T^i = STB(X^i)$
end
// Step2: LRM utilizes high-level features to extract
// Time-Reversal Semantic Asymmetry features
$T = LRM(T^1, T^2)$
// Step3: Deep supervision of the LRM output features
$M_{DS} = DeepSupervision(T)$
// Step4: Obtaining change mask using the Decoder
$M = Decoder(T)$

bidirectional convolutional long short-term memory networks (Bi-ConvLSTM) become a tool capable of more fully extracting their changed features. The Bi-ConvLSTM in the LRM effectively captures the vector nature of dual-temporal changes and aggregates bidirectional change features in a manner like an attention mechanism, making it more suitable for handling scenes with periodic changes, such as water bodies. Thus, introducing LRM with time-reversal semantic asymmetry as a tool for temporal feature extraction enhances the model's sensitivity and accuracy in capturing changes in such targets.

III. Method

Our model begins with multiple twin residual convolution blocks to extract spatial features from paired RS water body images. These spatial features are then fed into the STB to enrich their high-level semantic information. Subsequently, the features undergo processing in the LRM module, which leverages time-reversal semantic asymmetry to explore and aggregate spatiotemporal feature information related to changes. Within the encoder, we integrate the aggregated bidirectional temporal-spatial features with skip-connected multilayer differential spatial features. This integration ensures that during pixel-level prediction generation, the model can accurately account for the comprehensive impact of both spatial and temporal dimensions. Moreover, we enhance the capability of hidden layers to capture more representative temporal change features by incorporating DS into the LRM. The overall methodology of our TRSANet model is outlined in Algorithm 1.

TRSANet incorporates several key enhancements compared to other prominent change detection networks as follows:

1) The time-reversal semantic asymmetric LRM module is specifically tailored for water body change detection tasks. This module utilizes the Bi-ConvLSTM structure to capture bidirectional temporal features of changes and aggregates them efficiently through its aggregation module.
2) The Siam-Transformer module is integrated into the final layer of the spatial encoder. This addition aims to enhance the LRM module's ability to extract higher-level semantic features, thereby increasing sensitivity to spatial change features.
3) The integration of spatial and temporal feature extraction processes within the same encoder not only reduces the overall model parameter count but also decreases computational costs significantly.

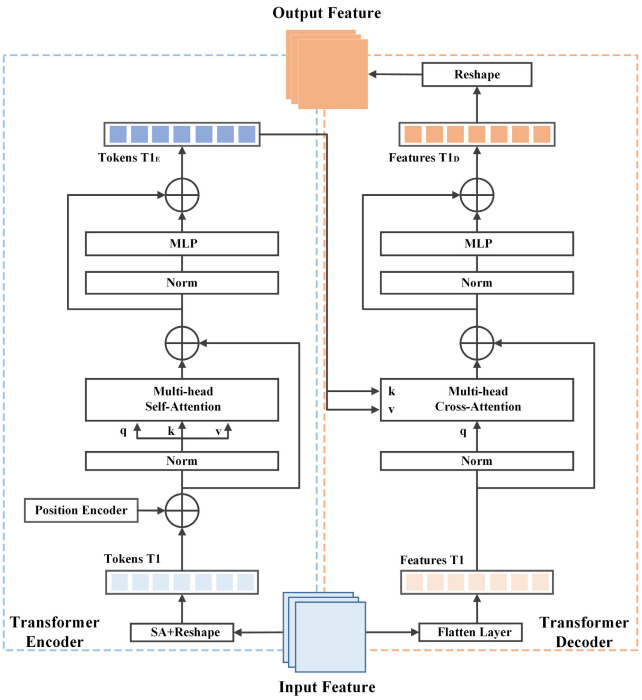


Fig. 2. Signal branch of the Siamese transformer block, the STB has two branches.

A. Extracting High-Level Semantic Features Using STB

The STB comprises a transformer encoder (TE) and a transformer decoder (TD), as illustrated in Fig. 2. The inspiration for the Siam-Transformer is derived from the BIT structure presented in [46]. Building upon this foundation, we have introduced modifications to Siam-Transformer, optimizing it further into the STB. The STB differs from BIT [46] by incorporating a Siamese TE and decoder structure designed to extract high-level semantic features from water body image pairs, thereby enhancing the LRM's ability to extract and integrate spatiotemporal features. In contrast, BIT employs only one TE and focuses on feature interaction between image pairs, aiming to accurately capture the feature information of the changing parts.

In contrast to the segmentation of images into patches and conversion into tokens as employed by vision transformer (VIT) [44] and Swim Transformer [45], STB adopts a different approach by transforming images into key semantic tokens. These tokens are then input into the TE for semantic feature enhancement, followed by pixel-space mapping in the TD. Specifically, we employ a 1×1 convolution and reshape operation to obtain the SA map MAP $\in R^{(L\times 1\times WH)}$ with the flattened input features $X_B \in R^{(1\times C\times WH)}$, and then sums over the final channel, obtaining input tokens $T^i \in R^{(L\times C)}$

$$T^i = \left(A^i\right)^T X_R^i = \left(\mathrm{SoftMax}\left(\mathrm{SA}\left(X_C^i; X_L^i\right)\right)\right)^T X_C^i \quad (1)$$

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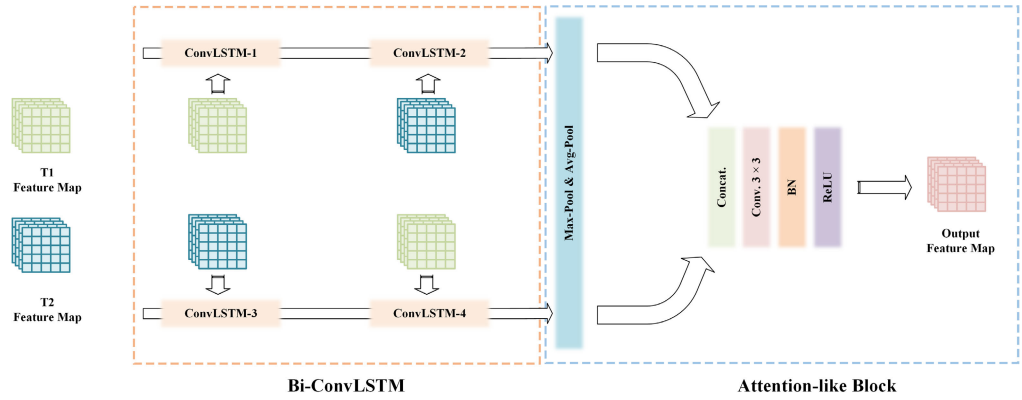


Fig. 3. Detailed structure of LRM, which includes the Bi-ConvLSTM and an attention-like module. The high-level semantic features of T1 and T2, provided by the STB, undergo a convolutional block individually. These features are then input into the Bi-ConvLSTM to extract bidirectional asymmetric spatiotemporal features. Following this, feature aggregation is carried out through attention-like block.

where $A^i \in R^{(HW\times L)}$ denotes the Attention Map, $X_R^i \in R^{(H\times W\times C)}$ represents the input features after the reshape operation. $\mathrm{SA}(X_C^i; X_L^i)$ represents the use of pointwise convolution to obtain the initial Attention Map $X_L^i \in R^{(H\times W\times L)}$ from the input feature $X_L^i \in R^{(H\times W\times L)}$.

The TE utilizes a multihead self-attention (MSA) mechanism to focus on information from different subspaces of the input tokens. This aids the model in acquiring global contextual information when processing images. Self-attention is the essence of MSA and the relationship between the input T^i and the components of self-attention (Query Q, Key K, Value V) is described as follows:

$$\begin{aligned} Q &= T^i W^q \\ K &= T^i W^k \\ V &= T^i W^v \end{aligned} \tag{2}$$

$$\mathrm{Att}(Q, K, V) = \mathrm{Softmax}\left(\frac{QK^T}{\sqrt{d}} + B\right)V. \tag{3}$$

In (2), $W^q, W^k, W^v \in R^{(C\times d)}$ are the learnable weights of the Linear Projection layers. Self-attention can be defined as shown in (3).

We adopt the same MSA module as ViT [44] and Swin-SUNet [18]. The output $\hat{T}^i \in R^{(L\times C)}$ from MSA undergoes layer normalization (LN), a multilayer perceptron (MLP) layer, and residual connections. This process results in the generation of tokens $T_E^i \in R^{(L\times C)}$ possessing compact high-level semantic information. The overall specific process of the TE module can be represented as shown in the following formula:

$$\begin{aligned} \hat{T}^i &= \mathrm{MSA}\big(\mathrm{LN}\big(T_E^i + E_p\big)\big) + T^i \\ T_E^i &= \mathrm{MLP}\big(\mathrm{LN}\big(\hat{T}^i\big)\big) + \hat{T}^i. \end{aligned} \tag{4}$$

Here, $\hat{T}^i$ represents the output from the MSA module, LN denotes LN, and MLP signifies the MLP layer. The Residual function incorporates the residual connection.

We need to use the TD to map the concepts learned by TE to the pixel space to obtain pixel-level features. We use the flattened features $X_F^i \in R^{(HW\times C)}$ as query (Q). We use the tokens T_E^i representing high-level semantic information generated by TE as key (K) and value (V). We compute multihead cross attention (MCA) and project it back to the pixel space to enhance feature information for the regions of interest. Finally, after the reshape operation, pixel-level high-level semantic features $X_D^i \in R^{(H\times W\times C)}$ are generated. The overall decoding process can be represented as shown in the following formula:

$$\begin{aligned} \hat{X}_D^i &= \mathrm{MCA}\big(\mathrm{LN}\big(X_F^i\big)\big) + X_D^i \\ X_D^i &= \mathrm{MLP}\big(\mathrm{LN}\big(\hat{X}_D^i\big)\big) + \hat{X}_D^i. \end{aligned} \tag{5}$$

Here, $\hat{X}_D^i$ represents the output of the MCA module, T_E^i represents the output of the TE module, X_F^i refers to the flattened features X_C^i, and X_D^i represents the output of the TD module.

B. Extract and Aggregate Bidirectional Asymmetric Spatiotemporal Features Using LRM

The core concept of the proposed TRSANet method is to exploit image pairs that capture asymmetric spatiotemporal features from their bidirectional change processes. To achieve this goal, a crucial step is to construct a module that imparts asymmetric semantics to the image pairs. While traditional RNN architectures are an option for handling temporal information, they are limited in their ability to process unidirectional data and are sensitive only to temporal information, making it challenging to capture spatial-temporal variations comprehensively. Considering this, we introduce LRM (LRM, as illustrated in Fig. 3). LRM employs Bi-ConvLSTM to compute the asymmetric spatiotemporal variations between the high-level semantic features extracted by the STB. The channel attention-like module is then utilized to aggregate bidirectional spatiotemporal variation features. By extracting and aggregating asymmetric bidirectional spatiotemporal semantic features, the model can better focus on essential spatiotemporal variations.

1) Bi-ConvLSTM: This architecture exhibits the characteristic of "time-reversal semantic asymmetry" primarily due to the application traits of LSTM in natural language processing. Natural language processing inherently possesses a certain temporal asymmetry: the current word is related not only to past words but also to future ones, thus capturing asymmetric semantic features. In LSTM, the processing of sequential data

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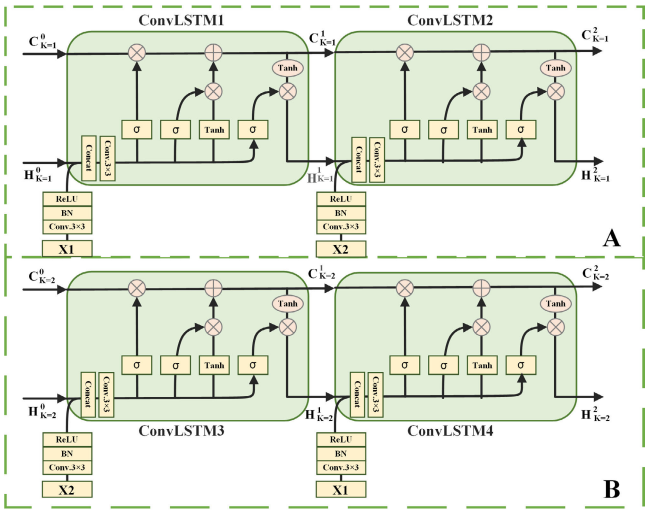


Fig. 4. Internal structure of Bi-ConvLSTM involves inputs X1 and X2, which are the high-level semantic features provided by STB for image pairs. The states $C_k^0, H_k^0 \in R^{(H\times W\times C)}$ are initialized as zero matrices. The extracted bidirectional asymmetric spatiotemporal features $H_{k=1,2}^2$ will be aggregate channel attention-like module.

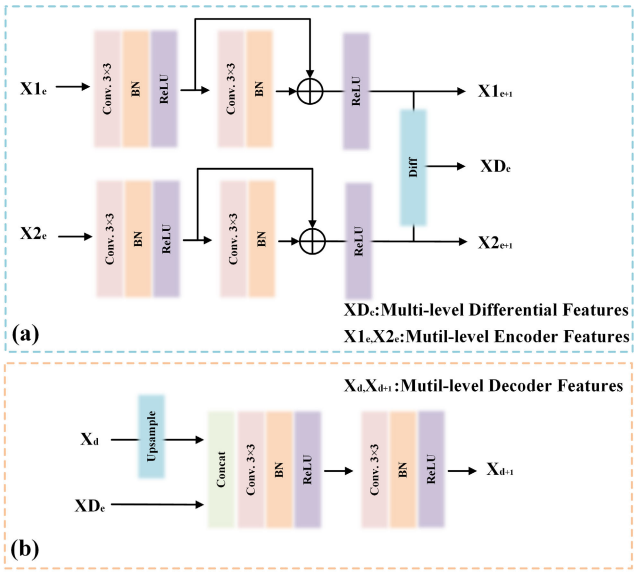


Fig. 5. (a) CNN backbone network and (b) decoder. It is noteworthy that in the last Res-block, there is no feature differencing operation; instead, it is directly connected to the STB module.

is controlled by input gates, output gates, and forget gates. The equations expressing the LSTM operations are as follows:

$$G_k^t = \phi\left(W_{G_k^t} \cdot \left[X_k^t, H_k^{t-1}\right]\right) \tag{6}$$

where G_k^t represents the forget, input, output, or cell gate of the LSTM for the k sequence at time step t, ϕ is the activation function such as sigmoid or tanh, $W_{G_k^t}$ denotes the weight matrix for the gate unit, and X_k^t, H_k^{t-1}, respectively denote the current input and the previous hidden layer.

Because of the limitation of traditional LSTM in computing temporal features for 1-D input time series and its inability to effectively utilize spatial features in 2-D space, particularly for high-dimensional data like RS image pairs, where matrix multiplication involves a significant computational cost, we extend the LSTM proposed in [47]. We introduce a convolution operation with a 3×3 kernel to replace the multiplication step in traditional LSTM. The schematic of Bi-ConvLSTM is illustrated in Fig. 4. The connection between ConvLSTM1 and ConvLSTM2, and the connection between ConvLSTM3 and ConvLSTM4 is sequential, forming LSTM chains A and B. However, the connection between these chains A and chains B is parallel and independent.

Specifically, the future state C_k^t, H_k^t of a particular cell are jointly determined by both temporal and spatial features. The update values for the forget gate, input gate, output gate, or cell gate of ConvLSTM $f_k^t, i_k^t, o_k^t, \tilde{C}_k^t \in R^{(H\times W\times C)}$ are given by

$$G_k^t = \phi\left(f^{3*3}\left(f_{\text{cat}}\left(X_k^t, H_k^{t-1}\right)\right)\right). \tag{7}$$

In the above equations, G_k^t represents the forget gate f_k^t, input gate i_k^t, output gate o_k^t, or the update value for the cell state $\tilde{C}_k^t$ at time step t for the k sequence in LSTM. The function ϕ represents activation functions like sigmoid or tanh. f^{3*3} denotes a convolutional layer with a kernel size of 3×3, and $f_{\text{cat}}(X_k^t, H_k^{t-1})$ represents the concatenation of the current input X_k^t and the previous hidden layer H_k^{t-1} along the channel dimension. The current forget gate, input gate, cell state, and hidden state $C_k^t, H_k^t \in R^{(H\times W\times C)}$ are determined by the following equation:

$$\begin{aligned} C_k^t &= f_k^t \odot C_k^{t-1} + i_k^t \odot \tilde{C}_k^t \\ H_k^t &= o_k^t \odot \tanh\left(C_k^t\right). \end{aligned} \tag{8}$$

Here, $\odot$ denotes the element-wise multiplication.

The Bi-ConvLSTM proposed by us consists of two branches of ConvLSTM with opposite time sequences. With the addition of convolutional layers, Bi-ConvLSTM can better handle high-level semantic features provided by STB in both spatial and temporal dimensions.

2) Attention-Like Module: It is depicted in Fig. 3(b). We initially utilize global max-pooling and global average-pooling to aggregate crucial information from bidirectional features. Further spatial-temporal aggregation is performed using a convolutional block on the obtained information. The output M can be represented as follows:

$$\begin{aligned} \hat{M} &= f_{\text{cat}}\left[\text{MaxPool}\left(H_{K=2}^2, H_{K=1}^2\right)\text{AvgPool}\left(H_{K=2}^2, H_{K=1}^2\right)\right] \\ M &= \text{ReLU}\left(\text{BN}\left(f^{3*3}\left(\hat{M}\right)\right)\right). \end{aligned} \tag{9}$$

C. Network Details

1) CNN Backbone: To effectively capture the intricate spatial characteristics, present in dual-temporal RS images, this study introduces a compact and efficient Siamese CNN encoder as the foundational network for TRSANet. The encoder architecture comprises four consecutive downsampling convolutional blocks, depicted in Fig. 5(a). To mitigate potential issues such as gradient vanishing or exploding gradients caused by the increased depth of the CNN, significant enhancements were made to the original two-layer convolutional blocks. These improvements include the incorporation of ReLU activation layers and batch normalization (BN) layers, which contribute to stabilizing the training

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TABLE I
DETAILED INFORMATION ABOUT THE SOURCES, PROPORTIONS, CATEGORY, AND LOCATIONS OF THE WATER-CD DATASET

Area	Pairs	Proportion	Category	Base
Lake of the Middle-Lower Yangtze	669	58.22%	Lake	China
George Lake	20	1.74%	Lake	Australia
Jomuna River	413	35.94%	River	Bangladesh&India

process. Moreover, to better preserve spatial details, a residual connection was integrated into the second convolutional block, thereby transforming it into a Res-Block.

2) Decoder: In order to better analyze the captured spatiotemporal change information, a simple and efficient decoder is essential. Similar to the CNN encoder backbone network, the decoder comprises four consecutive up-sampling convolutional blocks, depicted in Fig. 5(b), to progressively predict change information. The input to the decoder is the spatial-temporal aggregated feature (M) outputted by LRM. We integrate multilevel differencing features from the Siamese CNN backbone output into the decoding process. Post fusion, these features undergo analysis through two convolutional blocks equipped with ReLU and BN layers, and the resulting output is fed into the subsequent layer of up-sampling decoding convolutional blocks. Finally, the decoder generates a decoding feature map, which is transformed into a probability map via SoftMax and binarized using a threshold of 0.5 to derive the change image. Our experimental results indicate that a threshold of 0.5 optimally suits our dataset.

3) Loss Function: To address limitations in existing spatiotemporal change detection networks, which typically only supervise spatial information at the output layer, we extend supervision to the module responsible for extracting spatiotemporal features. DS [48] is introduced to mitigate the opacity and lack of supervision in training deep convolutional networks. DS involves incorporating side classifiers to directly supervise hidden layers, thereby enhancing the exploration of spatio-temporal features by our proposed LRM. We utilize a binary weighted cross-entropy loss function. To further mitigate gradient vanishing and bolster the targeted learning capacity of LRM, DS with $\alpha = 0.4$ is applied to the spatiotemporal features outputted by LRM. This approach augments the ability of hidden layers to capture more nuanced change features. The formulated loss function is expressed as follows:

$$l(P, R) = -\frac{1}{N_s}\sum_{ij}\left[\omega R_{ij}\log\left(P_{ij}\right) + (1-\omega)R_{ij}\log\left(1-P_{ij}\right)\right]$$

$$L = l(P_{\mathrm{NET}}, R) + \alpha l(P_{\mathrm{DS}}, R). \tag{10}$$

In this context, N_s refers to the number of samples, P_{NET} represents the predicted result of the network, P_{DS} represents the predicted result of the DS module, R represents the ground truth, and α is the weight factor for DS.

IV. EXPERIMENT

In this section, we validate the effectiveness and domain adaptability of our proposed method through experiments on two change detection datasets: Water-CD for water body change detection and LEVIR-CD for building change detection. Detailed introductions to these datasets and the evaluation metrics used are provided first. Subsequently, a series of comparative experiments assess the capabilities of the TRSANet network. Additionally, ablation studies are conducted on our proposed method to demonstrate its effectiveness specifically in water body change detection tasks.

A. Experimental Setup

Water-CD dataset serves the purpose of facilitating water body change detection, filling a gap in publicly available datasets dedicated to this task. It includes seasonal lakes from the Yangtze River Basin in China and various seasonal water bodies such as those from the Jumna River in South Asia, detailed in Table I. Utilizing Sentinel-2 satellite imagery, we obtained dual-temporal RS data at a 10 m resolution, comprising Red (R), Green (G), and near-infrared (NIR) bands from cloud-free images captured during both rainy and dry seasons spanning 2016–2022.

Given the computational demands of training DL networks on large images, we employed a cropping strategy with a stride of 0. Initially dividing the images into 512 × 512 patches, we segregated them into Train, Validation, and Test datasets to prevent data leakage between sets. This approach ensured that multitemporal images from the same location remained within the same dataset subset. The resulting cropped dataset consists of 1149 pairs of dual-temporal water body change detection images, each sized 512 × 512. Subsequently, after applying window cropping with a stride of 128 on the training set, we augmented its size to 11 240 images of 256 × 256 dimensions. The validation and test sets underwent window cropping with a stride of 0, resulting in 1416 and 1656 images, respectively. Examples of the Water-CD dataset are illustrated in Fig. 6(a)–(f). For further details on accessing the Water-CD dataset, including download instructions, please refer to https://github.com/ChogoriPeak/Water-CD.

LEVIR-CD is a large public dataset for building change detection. To investigate the cross-domain adaptability of the network, we conducted experiments utilizing the publicly available LEVIR-CD dataset for building change detection [40]. The dataset consists of 637 pairs of RS image pairs, each with a size of 1024 × 1024. Due to GPU memory constraints, we employed a cropping strategy consistent with (AMCA) Xu [49] and cropped the data into nonoverlapping images of size 256 × 256. This resulted in a total of 7120 training image pairs, 1024 validation image pairs, and 2048 test image pairs. Some examples of the LEVIR-CD dataset are shown in Fig. 6(g)–(i).

B. Evaluation Metrics

For binary classification tasks, samples are categorized into four classes based on the combination of true class and

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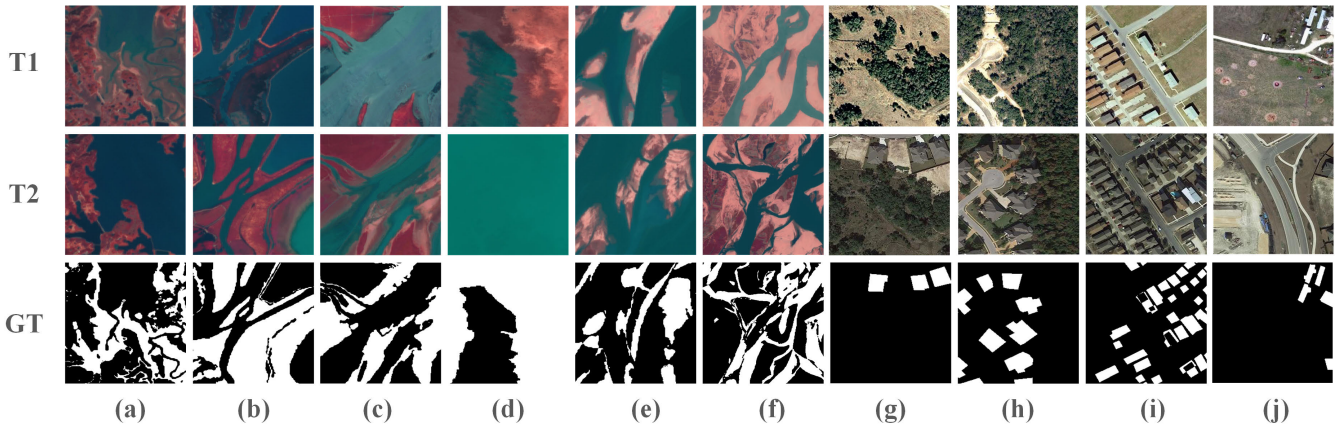


Fig. 6. Partial examples from two datasets. (a)–(c) Depict seasonal lakes in the middle and lower reaches of the Yangtze River, (d) is from Lake George, and (e) and (f) are from the Jomuna River. Examples (g)–(j) represent samples from the LEVIR-CD dataset, showcasing building updates and declines. Here, T1, T2, and GT denote images before the change, images after the change, and their corresponding ground truth, respectively.

model classification: true positive (TP), false positive (FP), true negative (TN), and false negative (FN). To comprehensively evaluate performance, we utilize five evaluation metrics, including Precision, Recall, $F1$ Score, overall accuracy (OA), and intersection over union (IOU). Specifically, Precision represents the ratio of correctly classified positive samples, while Recall denotes the ratio of correctly classified positive samples among the TP samples. The $F1$ Score is the harmonic mean of Precision and Recall. OA signifies the overall classification accuracy, and IOU measures the overlap between the predicted region and the ground truth region. Larger values of these metrics indicate better model performance, and these metrics can be expressed using the following formulas:

$$\begin{aligned} \text{Pre} &= \frac{\text{TP}}{\text{TP}+\text{FP}} \\ \text{Rec} &= \frac{\text{TP}}{\text{TP}+\text{FN}} \\ F1 &= \frac{2 * \text{Pre} * \text{Rec}}{\text{Pre}+\text{Rec}} \\ \text{OA} &= \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \\ \text{IoU} &= \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}}. \end{aligned} \tag{11}$$

C. Comparison to Other Methods

We compare TRSANet with seven recent state-of-the-art change detection methods to validate the effectiveness of our approach. The compared methods include CDNet [50], SNUNet [23], Spectral Feature Collaboration Network USSFCNet [51], Bidirectional Image Transformer BIT [45], Deep Supervised Image Fusion Network DSIFN [52], L-Unet [53], and P2V-CD [54]. A brief description of each method is as follows:

1) *CDNet:* Classic DL-based method.
2) *SNUNet:* U-shaped Siamese network with dense connections, using integrated attention modules to refine multilevel features. SNUNet-32, the most representative variant, was chosen for comparison.
3) *USSFCNet:* A lightweight spectral feature collaboration network based on multiscale feature fusion and attention mechanisms.
4) *BIT:* Employs a Transformer-based encoder to obtain high-level semantic features.
5) *DSIFN:* A deep supervised difference network with competitive accuracy on large-scale change detection datasets.
6) *L-Unet:* U-shaped Siamese convolutional network based on fully convolutional LSTM.
7) *P2V-CD:* A recent video-based method that transforms the pairs of dual-temporal images into pseudo-video frames and utilizes dual encoders for spatiotemporal feature extraction.

All experiments were implemented in PyTorch. For both datasets, the batch size was set to 8, and the Adam optimizer [55] was used as the optimization algorithm. The weight coefficient α for auxiliary loss was set to 0.4, and the weights for BCE loss were both 0.5. Data augmentation during training included random flipping, translation, and rotation in multiples of 90°. We carefully considered the different characteristics of each dataset and optimized hyperparameters to ensure convergence for all methods. For the Water-CD dataset, the learning rate was set to 0.008, trained for 45 epochs, and applied a step decay strategy with a decay rate and step size of 0.5 and 15, respectively. For the LEVIR-CD dataset, the learning rate was set to 0.008, trained for 100 epochs, and applied a step decay strategy with a decay rate and step size of 0.2 and 60, respectively.

The quantitative analysis of experimental results on the Water-CD dataset is presented in Table II, where the best values are highlighted in bold, and the second-best values are underlined. Compared to other methods, TRSANet achieves the best results with a significant advantage in water body change detection. This underscores the superiority and applicability of our method in the context of water body change detection.

Fig. 7 presents visual examples of the performance of each method on the Water-CD dataset. Rows 1 and 2 are dual-temporal changed data from lakes situated in the reaches

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TABLE II
QUANTITATIVE RESULTS OF DIFFERENT CHANGE DETECTION METHODS ON THE WATER-CD DATASET AND THE LEVIR-CD DATASET

	Water-CD					LEVIR-CD				
Model	Pre (%)	Rec (%)	F1 (%)	OA (%)	IoU (%)	Pre (%)	Rec (%)	F1 (%)	OA (%)	IoU (%)
CDNet	85.41	85.54	85.48	94.67	74.67	91.27	86.19	88.65	98.88	79.62
DSIFN	89.49	88.89	89.19	96.05	80.48	92.67	87.14	89.82	98.99	81.53
BIT	91.50	86.70	89.03	96.08	80.23	89.57	88.54	89.05	98.89	80.26
P2V-CD	88.80	90.24	89.51	96.12	81.02	92.44	**90.09**	91.25	99.12	83.91
L-UNet	91.17	89.04	90.09	96.41	81.96	91.88	89.28	90.56	99.05	82.75
USSFC-Net	88.44	88.64	88.54	95.79	79.43	92.26	88.98	90.52	99.06	82.79
SNUNet	89.32	**90.78**	90.04	96.32	81.89	92.24	89.83	91.02	99.10	83.51
TRSANet	**93.25**	90.22	**91.71**	**97.01**	**84.69**	**92.97**	89.95	**91.44**	**99.14**	**84.22**

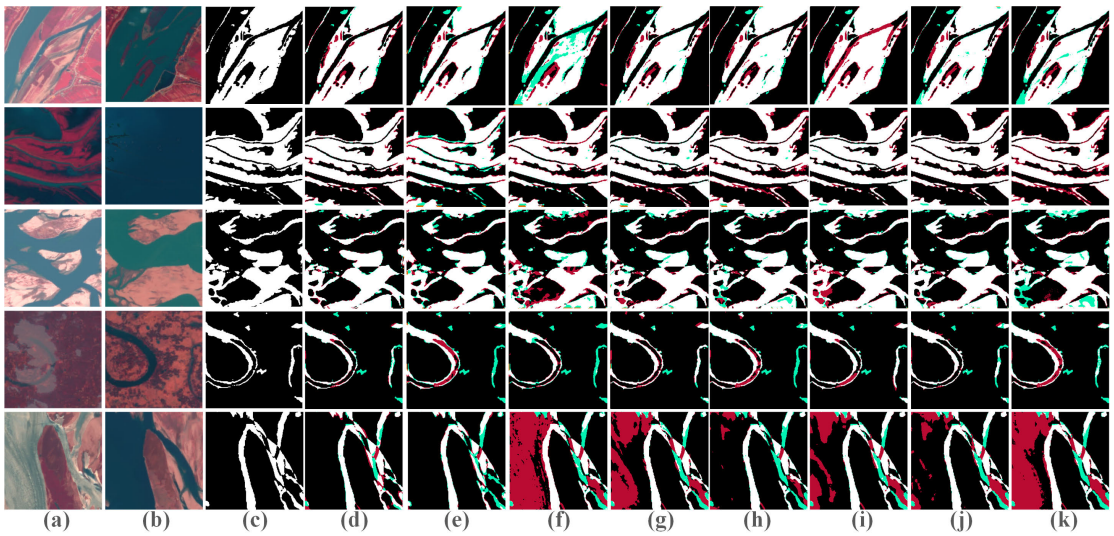


Fig. 7. Visual performance of different change detection methods on the water-CD dataset. (a)–(c) Represent images before and after the change, along with their ground truth. (d) TRSANet, (e) BIT, (f) CDNet, (g) DSIFN, (h) L-UNet, (i) P2V-CD, (j) SNUNet, and (k) USSFCNet. In our change maps, white pixels represent TP, black pixels represent TN, red pixels represent FP, and green pixels represent FN.

of the Yangtze River, respectively, while rows 3 and 4 depict river changes, with row 4 specifically showcasing an oxbow lake formed due to river channel diversion. Row 5 showcases seasonal pseudo-changes caused by increased sediment content in the water body. Visually, it is evident that the proposed TRSANet can adapt to a broader spectrum of water body changes and notably outshines the performance of competing models. By analyzing Fig. 7, we obtain the following conclusions:

1) TRSANet demonstrates remarkable precision in delineating the boundaries of changed regions. In the examples provided in rows 1, 2, and 3, most of the comparative approaches, including BIT, CDNet, and P2V-CD, tend to produce somewhat indistinct boundaries around the changed areas, primarily due to the presence of pseudo-changes, such as those caused by changes in riverbeds and tidal expanses. In contrast, our approach, as depicted in row d, not only mitigates the effects of these pseudo-changes but also delivers delineations of the substantial change zones. TRSANet (column d) is particularly adept at avoiding the influence of seasonal pseudo-changes. For instance, in row 5, column a, the image exhibits seasonal pseudo-changes caused by increased sediment content in water bodies. The detection results indicate that our model gets the best result and effectively counteracts seasonal pseudo-changes.
2) TRSANet excels in capturing changes in small water bodies. In the scenarios depicted in rows 2 and 3, which feature numerous minor aquatic bodies undergoing changes, several models like BIT, CDNet, DSIFN, L-UNet, P2V-CD, SNUNet, and USSFCNet, are observed to struggle with inaccuracies, including both FPs and missed detections, particularly in terms of defining the precise edges of minor altered areas. In contrast, TRSANet (column d) excels in capturing and restoring finer details.
3) TRSANet is more adept at avoiding the influence of clouds and fog compared to other methods. Cloud and fog interference is a significant issue in practical applications of water body change detection. The imagery in row 4 draws attention to a segment of the water body obscured by a veil of thin clouds. Apart from our method, alongside SNUNet and DSIFN, other techniques fail to accurately discern the changed contours hidden beneath cloud cover.

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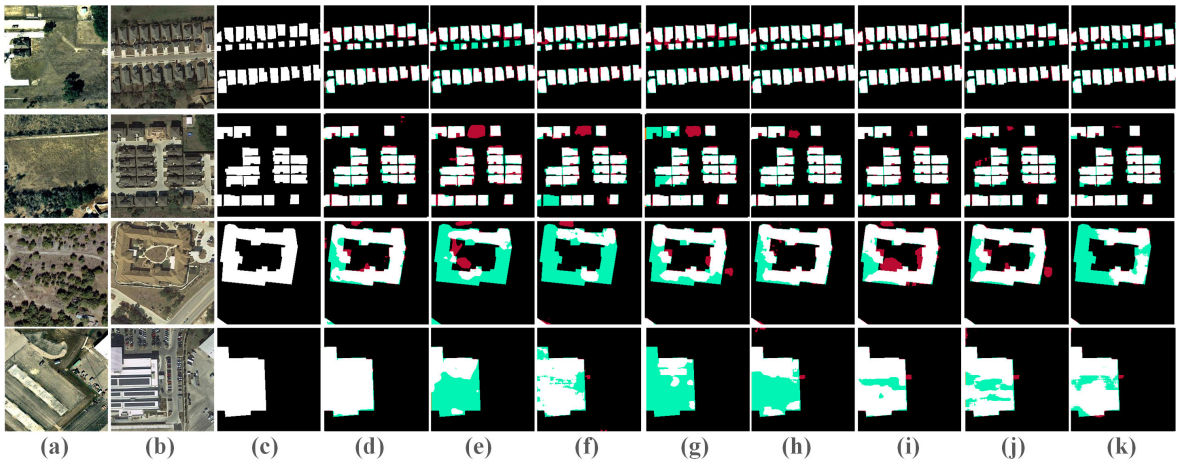


Fig. 8. Visual performance of different change detection methods on the LEVIR-CD dataset. (a)–(c) Represent images before and after the change, along with their ground truth. (d) TRSANet, (e) BIT, (f) CDNet, (g) DSIFN, (h) L-UNet, (i) P2V-CD, (j) SNUNet, and (k) USSFCNet. In our change maps, white pixels represent TP, black pixels represent TN, red pixels represent FP, and green pixels represent FN.

TABLE III
DIFFERENT ABLATION METHODS' IMPACTS ON THE MODEL

DIFF	LRM	STB	AUX	Pre (%)	Rec (%)	F1(%)	OA (%)	IOU (%)
X	X	X	X	89.64	88.09	88.86	95.95	79.95
√	X	X	X	91.84	88.12	89.94	96.38	81.72
√	√	X	X	92.21	90.15	91.17	96.80	83.77
√	√	√	X	**93.38**	89.66	91.48	96.94	84.30
√	√	X	√	91.21	89.37	90.28	96.47	82.28
√	X	√	√	92.47	88.59	90.49	96.58	82.63
√	√	√	√	93.25	**90.22**	**91.71**	**97.01**	**84.69**

Through the qualitative examination of the Water-CD dataset for water body change detection, our methodology exhibits outstanding performance in terms of accuracy, edge definition, and additional factors, surpassing competing methods in detecting alterations in the water body.

The quantitative analysis of experimental results on the LEVIR-CD dataset also is presented in Table II. It is noteworthy that LEVIR-CD, designed for building change detection, exhibits significant feature differences compared to the Water-CD water body change detection dataset. However, when compared to other change detection methods, our proposed TRSANet achieves the best results in building change detection. This demonstrates the cross-domain adaptation capability of our approach.

Fig. 8 displays visual examples of the performance of each method. Our proposed TRSANet demonstrates the ability to adapt to a more comprehensive range of building changes and achieves superior results compared to other models. From the analysis of Fig. 8, the following conclusions can be drawn:

1) TRSANet is notably proficient at identifying dense urban changes. In scenarios represented in the first and second rows, where the distribution of changing buildings is compact, and their scale is minimal, several methods (including BIT, CDNet, SNUNet) display varying levels of inaccuracies. Conversely, our approach effectively avoids the impact of pseudo-changes, resulting in enhanced edge clarity in changed portions.
2) TRSANet excels at capturing an expansive range of change areas. In cases illustrated in the third and fourth rows, concerning widely distributed changing buildings, competing methods struggle to comprehensively delineate the shape and borders of the entire area of change. Due to the high-level semantic features extracted by the STB and the extraction and aggregation of rich spatial-temporal features via the LRM module, our approach seamlessly integrates spatial and temporal features, resulting in the most precise predictions.

D. Ablation Studies

To further validate the effectiveness of the network model we proposed before, we conducted ablation experiments on the Water-CD dataset to demonstrate the efficacy of different modules in water body change detection. Under consistent experimental conditions, including the environment and parameter settings, we performed seven sets of ablation experiments by adding modules to our backbone. The quantitative results of all ablation experiments on the Water-CD dataset are presented in Table III. The "backbone" refers to the backbone network after removing all modules, and "DIFF" indicates the multilevel difference feature skip connections provided by the Siamese CNN backbone, where concatenation is used as a replacement when removed. "STB" represents the Siam-Transformer module, "LRM" represents the bidirectional asymmetric feature aggregation module proposed earlier, and "AUX" represents the depth supervision module.

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Overall, for water body change detection, no ablated model outperforms the complete TRSANet model in Table III. All modules complement each other and are indispensable. We conducted ablation experiments by enhancing and removing modules, starting from both the backbone and the complete TRSANet network. After replacing the concatenation operator with the difference feature operator in the skip connections of our backbone, our model gained a 1.08% increase in $F1$ score. With the addition of each module, our model achieved higher precision in the $F1$ score. Notably, after removing the LRM, there was a 1.22% loss in $F1$ scores, emphasizing the importance of LRM in extracting spatiotemporal features. Furthermore, removing the auxiliary branch for depth supervision resulted in a 0.23% decrease in $F1$ scores. Interestingly, after removing the STB module, we observed a negative impact on the depth supervision auxiliary branch. LRM's ability to handle spatiotemporal features diminishes, because without STB providing high-level semantic features. At this point, depth supervision on LRM has a counterproductive effect on training the hidden layers of the model, demonstrating a mutually complementary relationship among these three components.

V. Discussion

A. Effectiveness Evaluation of Time-Reversal Semantic Asymmetry

Experiments were carried out on the Water-CD dataset to validate the enhanced contribution of the proposed time-reversal semantic asymmetry method in detecting changes in water bodies, and compared to single-temporal feature extraction methods. Within the network, the Bi-ConvLSTM, responsible for generating time-reversal semantic asymmetry, was substituted with various modules. We temporarily refer to our model as TRSANet-LRM to facilitate comparison.

To investigate whether GRU also exhibits time-reversal semantic asymmetry, we modified GRU [42] into a structure like Bi-ConvLSTM, called Bi-ConvGRU, and developed a GRM analogous to LRM. The entire model is named TRSANet-GRM. To verify the impact of extracting only unidirectional variation features on the model, we converted the bidirectional structures to unidirectional ones, eliminating their time-reversal semantic asymmetry. These modified models are named TRSANet-ConvGRU(+)/(−) and TRSANet-ConvLSTM(+)/(−) respectively. ("+" denotes the forward changes, and "−" denotes the reverse one.) Additionally, the model called TRSANet-None is TRSANet-LRM with the LRM removed, meaning all temporal feature extraction capabilities are eliminated.

Metrics such as F1 Score, OA, IOU were measured TRSANet-LRM, TRSANet-ConvLSTM(+)/(−), TRSANet-GRM, TRSANet-ConvGRU(+)/(−), TRSANet-None models on the Water-CD dataset, as shown in Table IV. The results indicate that the performance of our proposed LRM is superior to other modules. This not only validates the efficacy of the time-reversal semantic asymmetry method but also demonstrates the effectiveness of the LRM.

Furthermore, the data show a slight advantage of TRSANet-LRM over TRSANet-GRM, a discrepancy attributed to the differences between Bi-ConvLSTM and Bi-ConvGRU. As mentioned in [42], for short sequences, TRSANet-GRM can achieve accuracy close to TRSANet-LRM with fewer parameters. Considering the acceptable loss in accuracy, selecting TRSANet-GRM for the extraction of bidirectional asymmetric spatiotemporal features in the TRSANet-LRM is regarded as a reasonable choice.

TABLE IV
Impact of the Time-Reversal Semantic Asymmetry Method on the Model

Area	F1(%)	OA(%)	IoU(%)
TRSANet-LRM	**91.71**	**97.01**	**84.69**
TRSANet-ConvLSTM+	90.83	96.69	83.20
TRSANet-ConvLSTM-	90.95	96.73	83.40
TRSANet-GRM	91.59	96.95	84.49
TRSANet-ConvGRU+	90.76	96.66	83.08
TRSANet-ConvGRU-	91.02	96.73	83.51
TRSANet-None	90.49	96.58	82.63

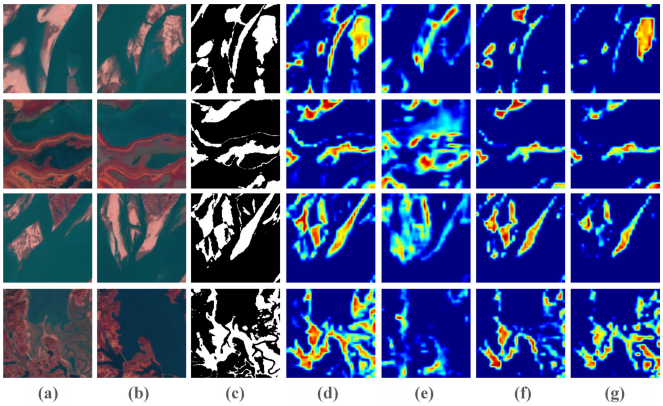


Fig. 9. Visual representation of features for different configurations of TRSANet: TRSANet-LRM with LRM, TRSANet-None without LRM, TRSANet-ConvLSTM (+) and TRSANet-ConvLSTM (−). ("+" denotes the forward changes, and "−" denotes the reverse one). (a)–(c) Displays images before and after the change, alongside the ground truth of the change areas. Panels (d)–(g) present TRSANet with LRM, TRSANet-None, and TRSANet-ConvLSTM (+), TRSANet-ConvLSTM (−), respectively. TRSANet-LRM can better focus on the regions of water body changes compared to the TRSANet-None and TRSANet-ConvLSTM (+), and TRSANet-ConvLSTM (−).

B. Visualization Feature of LRM

To better understand the LRM and time-reversal semantic asymmetry, we visualize the intermediate layer features of TRSANet-LRM and TRSANet-None (TRSANet-LRM without the LRM), TRSANet-ConvLSTM(+), and TRSANet-ConvLSTM(−) ("+" denotes the forward changes, and "−" denotes the reverse one.) by Grad-CAM [56], gradient-weighted class activation mapping (Grad-CAM) is a method used to visualize the decision-making process of neural networks. The experimental results, as shown in Fig. 9, reveal that in multiple pairs of dual-temporal water body change images, TRSANet-LRM can better focus on the regions of water body changes compared to the others. This demonstrates the effectiveness of LRM and time-reversal semantic asymmetry, a conclusion also supported by our experiments.

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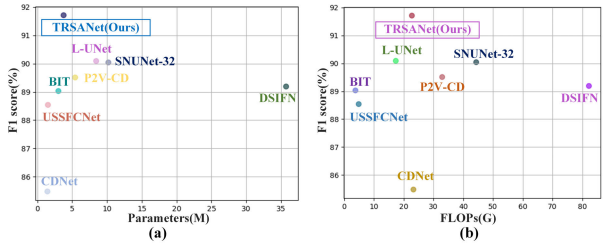


Fig. 10. (a) and (b) Comparison of computational efficiency (parameters and FLOPs) and accuracy ($F1$ Scores) among different methods is depicted, with our method positioned at the top, achieving the highest accuracy with a relatively low number of parameters and FLOPs.

TABLE V
IMPACT OF SUPERVISING THE MODEL WITH DIFFERENT WEIGHT FACTORS AND METHODS ON ACCURACY

α	Pre (%)	Rec (%)	F1 (%)	OA (%)	IoU (%)
LRM+0	**93.38**	89.66	91.48	96.94	84.30
LRM+0.1	92.35	90.36	91.34	96.86	84.07
LRM+0.2	92.40	**90.66**	91.52	96.92	84.37
LRM+0.3	92.57	90.23	91.39	96.88	84.14
LRM+0.4(ours)	93.25	90.22	**91.71**	**97.01**	**84.69**
LRM+0.5	92.84	90.08	91.44	96.90	84.23
LRM+0.6	92.20	90.39	91.29	96.83	83.97
STB+0.4	91.86	90.08	90.96	96.72	83.43

C. Parameter Analysis

In this section, we explore not only the visualization features of LRM and the efficacy of time-reversal semantic asymmetry but also the balance between the accuracy of our method and the efficiency of the network model. We evaluate model efficiency through two metrics: the number of parameters and floating-point operations per second (FLOPs). Fig. 10 presents data on the number of parameters and FLOPs across all methods. TRSANet has fewer parameters compared to most other models, while its FLOPs fall within a moderate range. Additionally, a comparison in Fig. 10 between the number of parameters and the $F1$ score indicates that TRSANet achieves the highest accuracy with few parameters.

D. Effectiveness Evaluation of DS on Models

To validate the impact of using different parameters and positions for depth supervision on TRSANet, we compared TRSANet models with different α values in the loss function and TRSANet with depth supervision on the STB. The comparison results are presented in Table V. When providing depth supervision to LRM with α set to 0.4, our model achieves the highest accuracy, surpassing the $F1$ value of TRSANet with depth supervision on the STB by 0.75%.

VI. CONCLUSION

In this article, we introduce a time-reversal semantic asymmetry method for detecting changes in water bodies. We developed a model named TRSANet, using our dedicated RS dataset for water body change detection, Water-CD. TRSANet excels in capturing and aggregating spatiotemporal features throughout the time inversion process, enabling the rapid and effective identification of changes. Specifically, the STB module, which incorporates a Transformer structure, adeptly extracts high-level semantic features from dual-temporal images, which are then utilized by the LRM module. The LRM leverages these features to efficiently discern temporal change information through a Bi-ConvLSTM.

We have validated the efficacy of our method on the water body change detection dataset and assessed its cross-domain adaptability with a building change detection dataset. The experimental outcomes suggest that our approach not only achieves superior accuracy but does so with a relatively modest parameter count. Furthermore, our proposed feature of time-reversal semantic asymmetry shows promising potential for integration into various methodologies that incorporate time feature extraction. A critical avenue for future research is to extend our model to multitemporal change detection tasks, which would enhance its ability to detect spatiotemporal changes, thereby increasing the accuracy of water body change detection. Additionally, this methodology could significantly contribute to the long-term monitoring and analysis of seasonal water bodies and assist in the detection and alerting of unusual changes in water bodies.

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